



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2018 – 0419



Agricultural Machinery – Coffee Pulper SUKI KOFI KING 1

Test Report Number	2018 – 0419
Valid Until	June 22, 2023
Agricultural Machinery	Coffee Pulper
Type	Cylindrical, Power-Operated
Brand	SUKI
Model	KOFI KING 1
Test Requested by	SUKI TRADING CORPORATION Agues, Ibabao, Lapu-lapu City

SUMMARY

Performance Criteria	Requirement as per PNS/PAES 252:2011	AMTEC Test
Pulping Efficiency, %, minimum	95.0	63.2
Pulping Recovery, %, minimum	93.5	92.5*
Losses %		
a. Separation Loss	1.0	0.13
b. Unpulped Loss	5.0	36.8
c. Scattering Loss	0.5	0
Purity, %, minimum	98	66.0
Mechanically Damaged Parchment Coffee, %, maximum	5.0	0
Noise Level maximum, dB(A), Feeder Level	92*	81.3

Note: *Computed as the ratio between the total weight of parchment coffee collected at the main outlet to the equivalent total weight of parchment coffee of the input coffee cherry to the machine, expressed in percentage

**Allowable noise level for six (6) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor, Philippines. 1983.

OBSERVATIONS

1. Operator's manual	Not provided
2. Previous valid test reports of similar brand and model	None
3. Prime mover used	0.75 kW AE Electric motor
4. Manufacturer	SUKI TRADING CORPORATION
5. Number of operators	Two (one for loading the input and one for collecting the output)
6. Collecting of outputs	The machine was provided with separate output chutes for the parchment coffee and pulps.
7. Transporting of machine	The machine has no transport wheels but can be manually transported.
8. Safety features	Belt-pulley transmission system was not provided with cover.
9. Failure or abnormalities that observed on the pulper or its components during and after pulping operation	No failure or abnormalities occurred.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date/s of Test	March 20, 2018
Location of Test	Agus, Ibabao, Lapu-lapu City
Items Tested	Input capacity, Output capacity, Pulping efficiency, Pulping recovery, Losses, Noise level, Power consumption, and Number of operators

METHODS OF TEST

The coffee pulper was tested based on the PNS/PAES 253:2011 Agricultural Machinery – Coffee Pulper – Methods of Test.

DESCRIPTION OF THE MACHINE

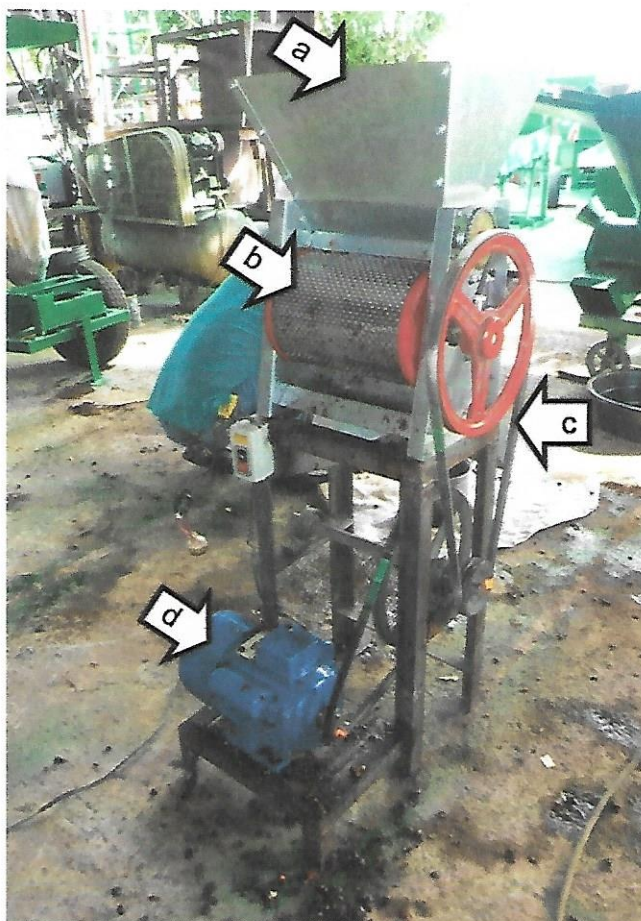


Figure 1. Parts of SUKI KOFI KING 1 Coffee Pulper powered by 0.75 kW AE Electric motor: (a) Hopper, (b) Pulping cylinder, (c) Belt-pulley transmission system, and (d) Electric motor.

MACHINE SPECIFICATIONS

Item	Manufacturer's Specification*	Verification by AMTEC
A.1 Main structure		
A.1.1 Overall dimensions (mm)		
A.1.1.1 Length	ND	715
A.1.1.2 Width	ND	440
A.1.1.3 Height	ND	1205
A.1.2 Weight, without the prime mover (kg),	ND	NM
A.2 Intake Hopper		
A.2.1 Bottom opening (L x W), mm	ND	280 x 55
A.2.2 Height from the ground, mm	ND	1205
A.2.3 Material	ND	G.I. Sheet
A.3 Feed Roller		
A.3.1 Size (L x D), mm	ND	279 x 45
A.3.2 No. of Vanes	ND	6
A.3.3 Material	ND	ND
A.4 Pulping Cylinder		
A.4.1 Type	ND	Cylindrical
A.4.2 Size (L x D), mm	ND	280 x 200
A.4.3 Material	ND	Stainless
A.5 Pulping Chamber Housing		
A.5.1 Material	ND	Cast Iron
A.6 Parchment coffee outlet		
A.6.1 Height from the ground, mm	ND	470
A.6.2 Material	ND	G. I. Sheet
A.7 Pulp outlet		
A.7.1 Height from the ground, mm	ND	NM
A.7.2 Material	ND	G. I. Sheet

Note: *Manufacturer's specifications were not supplied

ND – No Data; NM – Not measured

MACHINE SPECIFICATIONS (cont'n)

Item	Manufacturer's Specification*	Verification by AMTEC
A.8 Main Frame		
A.8.1 Material	ND	Angle Bar (38 mm 38 mm x 6 mm)
A.9 Power transmission system		
A.9.1 Electric Motor to Intermediate Shaft	ND	V-Belt and Pulley
A.9.1.1 Electric Motor Pulley	ND	51 mm x 1B x 24 mm
A.9.1.2 Intermediate Shaft Pulley	ND	305 mm x 1B x 25 mm
A.9.1.3 Belt	ND	A-47
A.9.2 Intermediate Shaft to Pulper Shaft	ND	V-Belt and Pulley
A.9.2.1 Intermediate Shaft Pulley	ND	102 mm x 1B x 25 mm
A.9.2.2 Pulper Cylinder Shaft Pulley	ND	305 mm x 1B x 18 mm
A.9.2.3 Belt	ND	B-59
A.9.3 Pulper Shaft to Feed Roller Shaft	ND	Chain and Sprocket
A.9.3.1 Pulper Shaft Sprocket	ND	15T, 40 mm Ø
A.9.3.2 Feed Roller Shaft Sprocket	ND	36T, 95 mm Ø
A.10 Electric Motor		
A.10.1 Brand	ND	AE
A.10.2 Type	ND	Capacitor-start induction motor
A.10.3 Make or manufacturer	ND	ABLE ELECTRIC CO., LTD.
A.10.4 Rated power, kW	ND	0.75
A.10.5 Rated speed, rpm	ND	1730
A.10.6 Phase	ND	Single
A.10.7 Voltage, V	ND	220
A.10.8 Current, A	ND	5.5
A.10.9 Frequency, Hz	ND	60

Note: *Manufacturer's specifications were not supplied

ND – No Data; NM – Not measured

RESULTS

Item		Data	
1	Crop Condition		
1.1	Variety	Robusta	
1.2	Source	Dawil, Negros Oriental	
1.3	Moisture content, %w.b.	70.4	
1.4	Coffee Cherry Dimensions, L x W x T, mm	15.6 x 11.6 x 9.8	
1.5	Parchment coffee to coffee cherry ratio	0.58	
1.6	Coffee cherry bulk density, kg/m ³	664.0	
2	Performance Test		
2.1	Speed of components, rpm	Without Load	With Load
2.1.1	Electric motor	1794	1790
2.1.2	Intermediate shaft	303	302
2.1.3	Pulping shaft	102	101
2.2	Noise level, dB(A)		
2.2.1	Feeder	75.4	81.3
2.2.2	Bagger	74.0	80.5
2.3	Weight of input, kg (coffee cherry)	10.82	
2.4	Weight of output, kg (parchment coffee)	8.84	
2.5	Input time, h	0.049	
2.6	Output time, h	0.050	
2.7	Input capacity, kg/h	220.8	
2.8	Output capacity, %	176.8	
2.9	Pulping efficiency, %	63.2	
2.10	Pulping recovery, %	92.5	
2.11	Losses, %		
2.11.1	Separation loss	0.13	
2.11.2	Unpulp loss	36.8	
2.11.3	Scattering loss	0	
2.12	Purity, %	66.0	
2.13	Mechanically damaged parchment coffee, %	0	
2.14	Electric Motor	Without Load	With Load
2.14.1	Power, kW	0.26	0.33
2.14.2	Current, A	1.53	1.68
2.14.3	Voltage, V	202	204

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SEP 12 2019
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